

Metabolic networks on PET-based C-Atlas for diagnosis of Alzheimer’s disease

Le Quoc Anh^{1,5,7}, Nguyen Tien Dung¹, Nguyen Phuong Trang¹, Pham Minh Tuan², Thanh Trung Le¹
Nguyen Thanh Trung⁴, Sara de Witte⁵, Mouloud Adel^{2,3}, Ismail Burak Parlak³, Karim Abed-Meraim⁷
Jeroen Van Schependom⁵, Sebastiaan Engelborghs^{5,6}, Guy Nagels^{5,6}, Nguyen Linh Trung^{1,*}

¹University of Engineering and Technology, Vietnam National University, Hanoi, Vietnam

²Aix-Marseille University, France, ³Galatasaray University, Turkey

⁴108 Military Central Hospital, Vietnam

⁵Vrije Universiteit Brussel, ⁶Universitair Ziekenhuis Brussel, Belgium

⁷PRISME Laboratory, University of Orleans, France

*Correspondence: linhtrung@vnu.edu.vn

Abstract—Alzheimer’s disease (AD) is the leading cause of dementia in elderly populations. Accurate and early diagnosis of AD has become an important objective of neuroimaging research. Among the available image modalities, ¹⁸F-fluorodeoxyglucose positron emission tomography (FDG-PET) is particularly well-suited to AD diagnosis. It has motivated the construction of metabolic networks that model AD as a disorder of interacting brain regions rather than isolated areas. The networks depend critically on the parcellation that defines their nodes. Yet, the atlases most commonly used in practice are derived from healthy anatomy or functional and do not reflect PET and AD specificity. In this paper, we propose a classification framework for AD detection that combines the PET-driven Coefficient-Atlas (C-Atlas) with kernel density estimate metabolic networks and a network-level region of interest (ROI) ranking. The proposed metabolic network pipeline achieves 93.1% accuracy for AD-vs-CN classification at $M = 300$ ROIs, surpassing the ROI-mean baseline. The proposed ROI ranking further boosts accuracy to 94.7% at only $K^* = 77$ regions. Within the same pipeline, C-Atlas achieves the highest classification accuracy among the four tested parcellations (AAL, Brainnetome, Schaefer, and C-Atlas) while requiring a more compact representation, confirming that C-Atlas provides a stronger basis for FDG-PET-based AD diagnosis.

Index Terms—Alzheimer’s disease, FDG-PET, metabolic network, brain atlas, support vector machine, ROI selection.

I. INTRODUCTION

Alzheimer’s disease (AD) is an irreversible neurodegenerative disorder and the leading cause of dementia in elderly populations. The World Alzheimer Report 2025 estimates that approximately 55 million people are currently living with dementia worldwide [1]. Because the underlying pathophysiology unfolds silently over years before clinical symptoms of AD emerge, accurate and early diagnosis has become an important objective of neuroimaging research.

Among neuroimaging modalities, ¹⁸F-fluorodeoxyglucose positron emission tomography (FDG-PET) has emerged as a highly sensitive tool for the early detection of AD-related metabolic alterations [2]. FDG-PET can reveal pathophysiological changes well before irreversible anatomical damage becomes visible on magnetic resonance imaging (MRI), and it

can quantitatively monitor cerebral glucose metabolism across disease stages [3]. FDG-PET is less dependent on neurovascular coupling, yields a more stable and reproducible signal at the single-subject level, and has therefore become a clinically established biomarker for AD diagnosis [4].

Beyond regional metabolic activity, much evidence indicates that AD is best understood as a network disorder [5]. The AD selectively disrupts large-scale systems of interacting brain regions rather than isolated areas. This perspective has motivated the construction of *metabolic networks* from FDG-PET data, where nodes correspond to brain regions and edges capture the statistical dependence of their glucose metabolic profiles [6].

A wide range of methodologies has been developed to estimate metabolic networks at the group level, including seed-based correlation analysis, independent and principal component analysis, sparse inverse covariance estimation, and graph-theoretic approaches that quantify the topology of the resulting networks through measures such as degree, strength, clustering coefficient, small-worldness, and modularity [3], [6]. Because static FDG-PET typically provides only a single image per subject, classical metabolic network analyses are restricted to group-level inferences. They cannot capture the individual variability required for personalized diagnosis. To overcome this limitation, recent work has proposed subject-level graph construction methods in which each region is represented either by its mean intensity (e.g., absolute-based and effect-size-based similarity) or by its full probability density function estimated through kernel density estimation (KDE) (e.g., Kullback-Leibler, dynamic time warping, and Wasserstein similarities) [7]. These individual metabolic networks preserve topological properties of group-level graphs while capturing subject-specific deviations, opening a path towards FDG-PET-based personalized network analysis for AD.

A critical ingredient of any brain-network pipeline is the choice of parcellation that defines the nodes. Predefined anatomical atlases such as the Automated Anatomical Labeling (AAL) atlas, the LONI Probabilistic Brain Atlas (LPBA40), and the Desikan–Killiany atlas, as well as more recent functional or cytoarchitectonic atlases such as Schaefer, Brainnetome,

This work was supported by the National Foundation for Science and Technology Development (NAFOSTED) of Vietnam under Grant 102.04–2021.55.

Human Connectome Project Multi-Modal Parcellation (HCP-MMP1.0), and Jülich, have been widely adopted in connectivity studies. Nevertheless, the vast majority of these atlases are derived from structural MRI or resting-state fMRI of healthy adults. Therefore, they neither reflect the metabolic organization captured by FDG-PET nor are aligned with the spatial signatures of AD-related neurodegeneration [8]. Applying MRI-derived parcellations to PET data can blur disease-relevant boundaries, mix regions with heterogeneous metabolic behavior, and ultimately attenuate the discriminative power of the resulting networks.

To address this mismatch, in this work, we propose to construct the metabolic network on a newly proposed FDG-PET based atlas called the Coefficient-Atlas (C-Atlas), with nodes defined by KDE, in order to enhance AD classification. To the best of our knowledge, the C-atlas is the first PET-driven, disease-adaptive brain atlas [8], [9]. Compared with the AAL atlas, C-Atlas has been shown to substantially improve AD classification accuracy, sensitivity, and specificity while remaining stable across data splits and spatial scales.

In the constructed network, each brain region is summarized by the distribution of its voxel intensities, and edges are obtained from pairwise similarity between distributions, via the Jensen–Shannon divergence (JSD) or the Wasserstein distance (WD). Experiments show that integrating a metabolic network with the C-Atlas improves AD classification performance compared to the C-Atlas alone.

In addition, to improve the use of the constructed metabolic network in AD classification (e.g., computational efficiency, focus on biologically relevant regions), we propose an efficient ranking strategy for regions-of-interest (ROI). This strategy preserves the most discriminative ROIs by introducing a pairwise score. We further evaluate the performance of the proposed ROI ranking method across multiple atlases, which again demonstrates the superiority of the metabolic network derived from the C-Atlas.

The remainder of this paper is organized as follows. Section II describes the dataset and the proposed framework, including the C-Atlas construction, the network construction, the ROI ranking, and the classification pipeline. Section III reports experimental results on the ADNI database, Section IV discusses our findings, and Section V concludes the paper.

II. MATERIALS AND METHODS

The overall proposed framework is illustrated in Fig. 1. Starting from the FDG-PET dataset, the two halves of the cohort are used for two distinct purposes: the first half is used to construct the C-Atlas and to estimate the ROI ranking, while the second half is reserved for performance evaluation.

A. Dataset and Preprocessing

In this study, we use ^{18}F -FDG PET data obtained from the ADNI database (adni.loni.usc.edu). Because our objective was AD diagnosis, only baseline scans were collected. For the AD class, we kept subjects who were diagnosed as AD at baseline and whose diagnostic label remained unchanged at

all subsequent follow-up visits. Likewise, for the cognitively normal (CN) class, we included subjects who were diagnosed as CN at baseline and remained CN throughout the follow-up period. Applying the above criteria yielded a final cohort of 598 subjects, whose demographic and clinical characteristics are summarized in Table I, where MMSE is Mini-Mental State Examination.

TABLE I
DATASET DEMOGRAPHIC

Characteristics	NC	AD
Number of subjects	295	303
Female/Male	161/134	124/179
Age (Mean $\pm$ std.)	72.97 $\pm$ 5.73	75.01 $\pm$ 8.12
MMSE (Mean $\pm$ std.)	29.08 $\pm$ 1.18	23.11 $\pm$ 2.29

For each selected subject, we collected the PET image already processed by ADNI according to the *Type-3* pipeline (*Co-registered, Averaged, Standardized Image and Voxel Size*), which yields volumes of size $160 \times 160 \times 96$ voxels at an isotropic resolution of 1.5 mm^3 . We then apply an additional preprocessing pipeline based on SPM12 to align all scans to a common anatomical space and normalize their intensities. First, each volume is manually reoriented so that its origin is aligned with the anterior commissure (AC), located at approximately (80, 89.5, 43.5) in voxel coordinates. The reoriented image is then warped into the Montreal Neurological Institute’s 152 brain template (MNI152) reference space; all parameters are kept at their default values, except the bounding box, which is reset to [90, 90, 108]. Next, an intensity normalization step is performed by dividing each voxel intensity by the mean cerebellar uptake value, which is commonly used as a reference because it is relatively spared in early AD. Finally, the resulting image is smoothed with an isotropic Gaussian kernel of full-width-at-half-maximum (FWHM) 8 mm to improve the signal-to-noise ratio and mitigate residual inter-subject anatomical variability.

B. C-Atlas Construction

We follow the C-Atlas construction method proposed in [8], which consists of two phases. In the first phase, PET images, along with their corresponding AD/CN labels, are used to train a linear support vector machine (SVM) classifier. Assume that we have N PET images available for constructing the C-Atlas. We use a brain mask extracted from the AAL atlas to select only the voxels lying inside the brain in the PET image, resulting in exactly 185 355 voxels. Stacking these vectors across subjects yields the training matrix $\mathbf{X}_{\text{train}} \in \mathbb{R}^{N \times 185\,355}$. A linear SVM is fitted on $\mathbf{X}_{\text{train}}$, yielding a coefficient vector $\mathbf{c} \in \mathbb{R}^{185\,355}$. The coefficient vector is then min–max normalized to the range $[0, 1]$, producing a normalized coefficient vector $\mathbf{c}_{\text{norm}} \in \mathbb{R}^{185\,355}$. This normalized vector is subsequently reshaped into a 3D volume, denoted by $\mathcal{W} \in \mathbb{R}^{91 \times 109 \times 91}$.

In the second phase, $\mathcal{W}$ is used as the input of the SLIC algorithm to delineate the brain regions. Specifically, we apply a mask-aware variant of the SLIC algorithm (maskSLIC) for

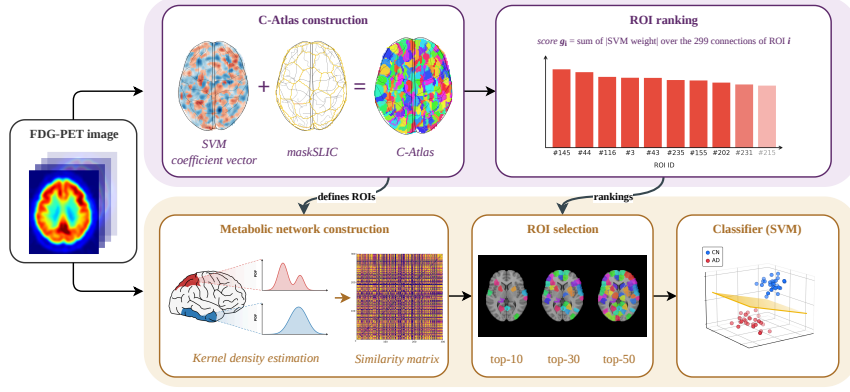


Fig. 1. Overview of the proposed framework for AD diagnosis from FDG-PET.

parcellation [10]. maskSLIC seeds cluster centers uniformly within the brain mask, then iteratively reassigns voxels to the nearest center based on a joint spatial and feature distance. The result is a set of spatially contiguous ROIs that respect the boundaries of the discriminative map. The SLIC algorithm exposes a parameter that determines the number of regions to extract. In this study, following the same multi-scale strategy as the Schaefer atlas, we vary the number of regions from 100 to 500 in steps of 100, yielding a family of C-Atlases at five different spatial scales.

C. Metabolic Network Construction

We construct metabolic networks from each PET image by following the pipeline proposed in [11], which consists of two successive phases: a *node-representation* phase, where each ROI is summarized by the probability density function (PDF) of its voxel intensities, and a *connectivity-representation* phase, where pairwise similarities between PDFs are converted into the edges of an undirected weighted graph.

In the *node-representation* phase, let R_i denote the set of voxel intensities lying inside the i -th ROI of a given subject. The underlying PDF p_i is estimated non-parametrically through a Gaussian kernel density estimate (KDE) over the values in R_i . The settings of this KDE depend on the similarity measure adopted in the next phase.

In the *connectivity-representation* phase, the metabolic relationship between any pair of ROIs i and j is quantified by a similarity measure $d(p_i, p_j)$ between their PDFs. In this work, we consider two measures, namely the JSD and the WD [11], each with its own KDE configuration. JSD uses a *fixed* KDE setting [12]. In contrast, the WD pipeline uses an *adaptive* KDE setting [11]. For both measures, the dissimilarity is ultimately mapped to an edge weight via an exponential transformation

$$s_{ij} = \exp(-d(p_i, p_j)), \quad (1)$$

where p_i, p_j are the KDE-based PDFs of ROIs i and j , $d(\cdot, \cdot)$ is either JSD or WD, and $s_{ij} \in [0, 1]$ is the resulting metabolic-similarity edge weight, which approaches 1 for strongly co-metabolic ROI pairs and 0 for ROI pairs with markedly different

metabolic profiles. Applying (1) to every pair of ROIs yields, for each subject, a symmetric similarity matrix $\mathbf{S} \in \mathbb{R}^{M \times M}$ with unit diagonal and off-diagonal entries in $[0, 1]$, where M is the number of ROIs of the parcellation. This matrix represents the metabolic network of the subject.

D. ROI Ranking

The similarity matrix $\mathbf{S}$ might contain several ROI pairs that contribute little to the AD-vs-CN distinction. We introduce a strategy for quantifying the importance of each ROI directly at the network level, hereafter called the *pairwise score* g_i , which is built from the similarity matrices.

The pairwise score is computed on the same set of N PET images that is used in the C-Atlas construction phase. For a parcellation with M ROIs, the network-construction stage produces, for each PET image, a similarity matrix $\mathbf{S} \in \mathbb{R}^{M \times M}$. Because $\mathbf{S}$ is symmetric with unit diagonal, all of its discriminative information is contained in its upper triangle, which yields $L = M(M - 1)/2$ scalar features per image. Stacking these feature vectors across the N training images gives a matrix $\mathbf{E}_{\text{train}} \in \mathbb{R}^{N \times L}$. A linear SVM classifier is then fitted on $\mathbf{E}_{\text{train}}$ with the corresponding AD/CN labels, yielding a coefficient vector $\mathbf{w} \in \mathbb{R}^L$. Each entry of $\mathbf{w}$ is associated with a specific ROI pair (i, j) ; arranging these coefficients into a symmetric matrix $\mathbf{W} \in \mathbb{R}^{M \times M}$ (with $\mathbf{W}_{ij} = \mathbf{W}_{ji}$ and $\mathbf{W}_{ii} = 0$), the pairwise score of ROI i is finally defined as

$$g_i = \sum_{j \neq i} |\mathbf{W}_{ij}|, \quad i = 1, \dots, M, \quad (2)$$

that is, the total contribution of all edges incident to ROI i to the AD-vs-CN distinction. Because the score depends on the choice of similarity measure used to build $\mathbf{S}$ in (1), two variants of g_i are computed, one per choice of $d(\cdot, \cdot)$. Both rankings are estimated once on the training set and then used as fixed ROI selectors at evaluation time.

Top- K ROI selection. For a chosen budget $K \leq M$, we keep only the K highest-scoring ROIs according to g_i and *rebuild* the similarity matrix from scratch on this subset, yielding a new $K \times K$ submatrix per subject.

E. Classification and Evaluation

Since the C-Atlas and the pairwise score g_i are both learned on the same N training subjects, the remaining $D = 598 - N$ subjects of the cohort introduced in Section II-A are reserved for evaluation. For each of these D test subjects, the network-construction pipeline produces a similarity matrix $\mathbf{S}$, from which we extract L scalar features (the upper-triangular entries described in Section II-D). These feature vectors are then fed into a classifier that distinguishes between AD and CN. We assess classification performance using three standard metrics: accuracy (ACC), sensitivity (SEN), and specificity (SPE).

III. EXPERIMENTS AND RESULTS

A. Data split and Cross-validation

The cohort of 598 subjects described in Section II-A is randomly split into two halves, stratified by class label. The first half is used for C-Atlas construction (Section II-B) and for the estimation of the pairwise score g_i (Section II-D), while the second half is reserved for the evaluation of the classification. After the split, each half contains exactly 299 subjects. On the evaluation set, classification performance is assessed by a 10-fold cross-validation procedure repeated 20 times with different random fold partitions, which yields 200 independent train/test fits per configuration; we report the mean and standard deviation of the three metrics (ACC, SEN, SPE) across these 200 fits to ensure the reliability of the reported results.

B. AD Classification Performance

We evaluate the AD classification performance of the metabolic network pipeline on the family of C-Atlases. We consider three feature representations: the ROI-wise *Mean value* baseline of [8], and two variants of the metabolic network built from the JSD and WD similarity measures, respectively. Each representation is fed into an SVM classifier with two kernels, linear and radial basis function (RBF). All pipelines are evaluated on the same five C-Atlas scales (100, 200, 300, 400, and 500) produced in Section II-B.

The results of the six pipelines across the five C-Atlas scales are reported in Table II. The metabolic network pipelines generally achieved higher performance than the baseline pipelines of [8]. The highest overall accuracy of 93.1% is obtained by using the JSD-based metabolic network with $M = 300$ and an RBF kernel for the SVM classifier.

C. ROI Ranking

Building on the previous experiment, C-Atlas with $M = 300$ regions was identified as a suitable atlas for AD classification. We now use this fixed parcellation to investigate how informative each region is for the AD-vs-CN distinction and whether the classification performance can be improved by restricting the analysis to a small subset of the most discriminative regions.

We compare two ROI-ranking strategies, each producing a score for each region of the C-Atlas. The first strategy is the voxel-level coefficient score v_i proposed in [8], which re-uses the linear SVM coefficient vector $\mathbf{c} \in \mathbb{R}^{185 \times 355}$ (Section II-B).

TABLE II
PERFORMANCE ON AD CLASSIFICATION TASK USING C-ATLAS

Method	# ROIs	ACC	SEN	SPE
Mean value linear [8]	100	88.5 $\pm$ 5.4	87.0 $\pm$ 8.1	90.1 $\pm$ 7.9
	200	90.5 $\pm$ 5.0	89.2 $\pm$ 7.8	91.9 $\pm$ 6.9
	300	92.3 $\pm$ 5.0	90.4 $\pm$ 7.9	94.2 $\pm$ 6.1
	400	91.8 $\pm$ 5.1	89.6 $\pm$ 7.9	94.0 $\pm$ 6.5
	500	91.3 $\pm$ 4.9	89.1 $\pm$ 7.9	93.6 $\pm$ 6.3
Mean value RBF	100	88.4 $\pm$ 5.6	87.9 $\pm$ 8.4	88.9 $\pm$ 8.0
	200	88.7 $\pm$ 5.5	89.0 $\pm$ 8.1	88.5 $\pm$ 8.1
	300	88.5 $\pm$ 5.5	88.7 $\pm$ 8.1	88.2 $\pm$ 8.1
	400	89.2 $\pm$ 5.3	89.6 $\pm$ 7.7	88.7 $\pm$ 7.9
	500	89.0 $\pm$ 5.3	89.2 $\pm$ 7.8	88.8 $\pm$ 7.9
Network JSD linear	100	90.1 $\pm$ 5.0	90.1 $\pm$ 7.8	90.1 $\pm$ 7.1
	200	91.1 $\pm$ 5.4	89.8 $\pm$ 8.2	92.5 $\pm$ 6.8
	300	91.6 $\pm$ 5.1	90.3 $\pm$ 7.7	93.0 $\pm$ 6.7
	400	91.9 $\pm$ 5.2	91.3 $\pm$ 7.5	92.5 $\pm$ 6.4
	500	92.5 $\pm$ 5.1	91.5 $\pm$ 7.2	93.5 $\pm$ 6.3
Network JSD RBF	100	91.5 $\pm$ 5.3	91.6 $\pm$ 7.5	91.3 $\pm$ 6.8
	200	92.5 $\pm$ 4.5	93.3 $\pm$ 6.2	91.7 $\pm$ 6.4
	300	93.1 $\pm$ 4.4	93.8 $\pm$ 6.1	92.5 $\pm$ 6.2
	400	92.8 $\pm$ 4.4	93.5 $\pm$ 6.2	92.0 $\pm$ 6.4
	500	92.6 $\pm$ 4.6	93.4 $\pm$ 6.2	91.8 $\pm$ 6.5
Network WD linear	100	88.9 $\pm$ 5.1	88.7 $\pm$ 7.9	89.1 $\pm$ 7.6
	200	89.4 $\pm$ 5.5	89.0 $\pm$ 8.3	89.8 $\pm$ 7.8
	300	91.0 $\pm$ 5.1	90.1 $\pm$ 7.6	92.0 $\pm$ 6.4
	400	90.5 $\pm$ 5.4	89.8 $\pm$ 7.9	91.3 $\pm$ 7.2
	500	90.6 $\pm$ 5.2	90.2 $\pm$ 7.4	91.0 $\pm$ 7.3
Network WD RBF	100	91.8 $\pm$ 5.1	93.9 $\pm$ 6.2	89.6 $\pm$ 8.4
	200	91.8 $\pm$ 5.1	93.9 $\pm$ 6.1	89.7 $\pm$ 7.9
	300	91.8 $\pm$ 4.9	93.6 $\pm$ 6.1	90.1 $\pm$ 7.9
	400	91.9 $\pm$ 5.0	93.8 $\pm$ 6.1	90.0 $\pm$ 8.0
	500	92.1 $\pm$ 4.7	93.8 $\pm$ 6.1	90.4 $\pm$ 7.4

Note: “Mean value” and “Network” denote the feature type for classifier; **Bold** values indicate the highest score within a single pipeline, while values in **bold red** indicate the highest score across all pipelines.

For each ROI i , the score $v_i = \sum_{k \in R_i} |c_k|$ is defined as the sum of the absolute SVM weights of the voxels belonging to R_i . The second strategy is the pairwise score g_i introduced in Section II-D, which is computed directly at the network level of the similarity matrices $\mathbf{S}$. Both rankings are estimated on the same training half of the cohort defined in Section III-A, and the resulting ROI orderings are then used as fixed selectors at evaluation time.

For each ranking, we apply a top- K ROI selection: only the K highest-scoring ROIs of C-Atlas are kept, and the similarity matrix is *rebuilt* from scratch on this subset to yield a $K \times K$ submatrix per subject. Because the choice of similarity measure and SVM kernel also affects the downstream classifier, the top- K analysis is carried out for the four $\{\text{JSD}, \text{WD}\} \times \{\text{linear}, \text{RBF}\}$ configurations of the metabolic network pipeline, giving eight (ranking $\times$ kernel $\times$ measure) combinations in total.

Fig. 2 reports the top- K accuracy curves of the eight combinations on C-Atlas ($M = 300$). The proposed pairwise score g_i achieved higher accuracy than the voxel-level score v_i across all four kernel/measure pairs, with the largest gap observed when $K < 100$. Among the kernel/measure pairs, JSD performs better than WD for both kernels, and the RBF kernel improves

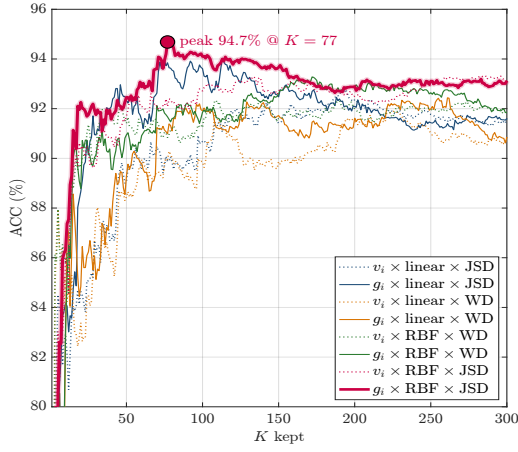


Fig. 2. Accuracy curves on C-Atlas ($M = 300$) when we use Top- K importance regions for AD classification. For readability, the y -axis is cropped to $[80\%, 96\%]$. Note that for $K < 10$, the classification accuracy of all eight pipelines remains below 70%.

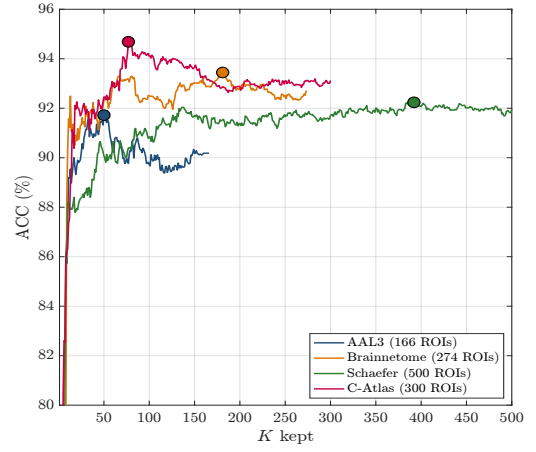


Fig. 3. Top- K accuracy curves of the four brain atlases (AAL3, Brainnetome, C-Atlas, and Schaefer). The filled dot on each curve marks the peak accuracy of the corresponding atlas.

over its linear counterpart for both measures. The best overall curve is obtained by combining the pairwise ranking g_i with the RBF kernel and the JSD measure, achieving a peak accuracy of 94.7% at $K = 77$. All curves quickly plateau after $K \approx 50$.

D. Atlas Comparison

In this section, we investigate whether the gain from C-Atlas is preserved when the parcellation is replaced with other widely used brain atlases within the same pipeline. We compare C-Atlas at $M = 300$ with three reference atlases that span the main families used in the connectivity literature: AAL3 with 166 anatomically defined ROIs, the Brainnetome atlas with 274 connectivity-derived ROIs, and the Schaefer atlas with 500 functional ROIs [11]. For each of the atlases, the KDE-based PDFs are computed on the ROIs of the atlas, the edges of the similarity matrix $\mathbf{S}$ are obtained from the JSD measure with the fixed KDE setting, the pairwise score g_i is recomputed independently on the training half of the cohort defined in Section III-A, and a top- K ROI selection is then applied. Classification is performed with an RBF SVM. In this experiment, K is swept from 1 up to the size M of each atlas.

The resulting top- K accuracy curves are shown in Fig. 3. The C-Atlas achieves the highest peak accuracy, exceeding those of the three reference atlases: Brainnetome (93.5%), Schaefer (92.2%), and AAL3 (91.7%). The peak of C-Atlas is reached with far fewer regions than the connectivity and function based atlases ($K^* = 77$ for C-Atlas versus 181 for Brainnetome and 392 for Schaefer). AAL3 requires the fewest regions ($K^* = 50$) to achieve its peak accuracy, but at a substantially lower level than the other three atlases.

IV. DISCUSSION

The results reported in Table II show that the proposed metabolic network pipelines generally outperform the baseline method in [8]. This result indicates that summarizing each ROI by the full PDF of its voxel intensities and exploiting pairwise

PDF similarities preserves more AD-discriminative information than the mean alone. The choice of similarity measure and SVM kernel also has a clear and consistent effect. Across all scales, JSD yields higher classification metrics than WD, and the RBF kernel consistently outperforms the linear kernel in network-based pipelines. The best configuration (Network $\times$ JSD $\times$ RBF) also exhibits the smallest standard deviations, suggesting that it is not only the most accurate but also the most stable. Furthermore, the influence of the number of regions M depends on the pipeline. The baseline method and the best configuration both peak at $M = 300$ and degrade for larger scales, whereas the linear-kernel pipelines tend to improve monotonically up to $M = 500$. The Network $\times$ WD $\times$ RBF pipeline is essentially flat across the whole range. Taken together, these observations suggest that $M = 300$ provides a favorable trade-off between representation compactness and classification accuracy. It is large enough to capture the spatial heterogeneity of AD-related metabolic alterations, yet small enough to keep the feature vector size and the risk of overfitting under control.

The top- K curves of Fig. 2 show that the proposed pairwise score g_i consistently outperforms the voxel-level baseline v_i across all four (kernel, measure) combinations. This advantage can be attributed to representational alignment. g_i aggregates the coefficients of a classifier trained on the same pairwise feature space that the downstream SVM operates on, so its scores directly reflect how each ROI contributes to the AD-vs-CN distinction, whereas v_i is one step removed because it aggregates voxel-level weights. Another observation is that the peak accuracy is reached at $K^* = 77$, i.e. with only about a quarter of the $M = 300$ ROIs of C-Atlas, and exceeds the accuracy obtained when the full atlas is used as input to the classifier. This observation indicates that a large fraction of the ROI pairs in $\mathbf{S}$ are uninformative or even detrimental for classification, and that the proposed ranking can effectively isolate the discriminative subset. Beyond the

peak, all curves rapidly plateau at $K \approx 50$, suggesting that the AD-discriminative metabolic signal, while distributed across multiple regions, is concentrated in a relatively small subset of top-ranked ROIs rather than uniformly spread across the whole brain.

The atlas comparison of Fig. 3 provides converging evidence that the choice of parcellation has a substantial impact on the discriminative power of the resulting metabolic network. Under the same pipeline (g_i , RBF, JSD), C-Atlas reaches a peak accuracy of 94.7%, above Brainnetome (93.5%), Schaefer (92.2%), and AAL3 (91.7%), and its curve dominates the three reference atlases over almost the entire K range. This advantage can be explained by the way C-Atlas is constructed: its ROI boundaries are explicitly learned to align with the gradient of voxel-level discriminative weights between AD and CN, so that each region is, by design, more informative for the target task than the anatomically, connectivity or function driven parcellations. Equally importantly, C-Atlas reaches its peak with $K^* = 77$ regions, which is much smaller than the peak budgets of Brainnetome ($K^* = 181$) and Schaefer ($K^* = 392$), and shows that the disease-adaptive parcellation also produces a more compact representation. AAL3 reaches its peak with even fewer regions ($K^* = 50$), but at a substantially lower accuracy, suggesting that its coarse anatomical boundaries quickly saturate the limited information they can carry rather than reflecting genuine compactness.

This study has several limitations. First, both the C-Atlas and the pairwise score g_i are supervised: they must be re-learned whenever the task, the cohort, or the imaging protocol changes, and care must be taken to keep their construction strictly inside the training fold. We mitigated leakage by deriving them exclusively on the 299 training subjects, but the supervised nature of the pipeline still limits its direct portability. Second, the entire experimental protocol relies on the ADNI database with a single binary task (AD vs. CN). The generalizability of our findings to other cohorts, to early-stage subjects such as mild cognitive impairment (MCI), and to other neurodegenerative diseases remains to be established. Finally, while the pairwise score g_i effectively identifies a discriminative subset of ROIs, the analysis remains purely predictive. Connecting the top-ranked ROIs to the spatial signatures of AD-related hypometabolism and to known disease-affected networks would strengthen the clinical interpretability of the framework and is a next step.

V. CONCLUSION

We proposed a framework combining the PET-driven C-Atlas with KDE-based metabolic networks and a network-level ROI ranking for the diagnosis of Alzheimer’s disease from FDG-PET. Our experiments on the ADNI database showed that the proposed framework improves classification performance over the ROI-mean baseline. The disease-driven C-Atlas provides a more compact and discriminative parcellation than widely used anatomical, connectivity and function based atlases. Future work will extend the framework to other neurodegenerative conditions and investigate the spatial interpretability of the most discriminative regions.

REFERENCES

- [1] L. Clare and Y.-H. Jeon, “World Alzheimer report 2025: Reimagining life with dementia – the power of rehabilitation,” Alzheimer’s Disease International, London, England, Tech. Rep., Sep. 2025.
- [2] A. Verger and E. Guedj, “The renaissance of functional ^{18}F -FDG PET brain activation imaging,” *European Journal of Nuclear Medicine and Molecular Imaging*, vol. 45, no. 13, 2018.
- [3] X. Pan, M. Adel, C. Fossati, T. Gaidon, and E. Guedj, “Multilevel feature representation of FDG-PET brain images for diagnosing Alzheimer’s disease,” *IEEE Journal of Biomedical and Health Informatics*, vol. 23, no. 4, 2019.
- [4] M. Wang et al., “Individual brain metabolic connectome indicator based on kullback-leibler divergence similarity estimation predicts progression from mild cognitive impairment to Alzheimer’s dementia,” *European journal of nuclear medicine and molecular imaging*, vol. 47, no. 12, 2020.
- [5] P. Baloni et al., “Metabolic network analysis reveals altered bile acid synthesis and metabolism in Alzheimer’s disease,” *Cell Reports Medicine*, vol. 1, no. 8, 2020.
- [6] T. Horowitz, M. Doyen, S. P. Caminiti, I. Yakushev, A. Verger, and E. Guedj, “Metabolic brain PET connectivity,” *PET Clinics*, vol. 20, no. 1, 2025.
- [7] M. T. Pham, T. Horowitz, M. Adel, J. Wojak, L. T. Nguyen, and E. Guedj, “Comparative evaluation of graph construction methods for individual brain metabolic network from FDG-PET images: An ADNI study in healthy subjects,” *European Journal of Nuclear Medicine and Molecular Imaging*, vol. 53, 2026.
- [8] M. T. Pham, T.-L. Phan, M. Adel, S. Bourennane, L. T. Nguyen, and E. Guedj, “C-Atlas: A brain mapping based on FDG-PET images for Alzheimer’s disease diagnosis,” in *2022 RIVF Int. Conf. Computing and Communication Technologies (RIVF)*, 2022.
- [9] P. M. Tuan et al., “Novel PET-driven brain mapping for diagnosis of Alzheimer’s disease: Integrated feature extraction and ranking capabilities,” *APSIPA Transactions on Signal and Information Processing*, Jun. 2026, [accepted].
- [10] B. Irving, “maskSLIC: Regional superpixel generation with application to local pathology characterisation in medical images,” *arXiv:1606.09518*, 2016.
- [11] M. T. Pham, M. Adel, L. T. Nguyen, and E. Guedj, “Wasserstein-based distance for constructing multi-scale individual brain networks from FDG-PET images: Application to Alzheimer’s disease diagnosis,” in *Proc. 32nd European Signal Processing Conf. (EUSIPCO)*, 2024.
- [12] W. Li, Y. Tang, L. Peng, Z. Wang, S. Hu, and X. Gao, “The reconfiguration pattern of individual brain metabolic connectome for Parkinson’s disease identification,” *MedComm*, vol. 4, no. 4, 2023.